Seunghyeon Lee (이승현 · Clay Lee)

Ph.D. Candidate (expected Aug 2026) · Forest Structure & Remote Sensing (LiDAR / GEDI) · Seoul National University

shlee5598@snu.ac.kr | +82-10-8455-5598 | drseunghyeonlee.com | ORCID 0000-0003-4612-065X | Google Scholar

EDUCATION

Ph.D. Candidate, Interdisciplinary Program in Landscape Architecture, Seoul National University — 2022.03 — Aug 2026 (expected)
Dissertation: "LiDAR-Based Characterization of Vertical Structure and Disturbance in Temperate Forests" · Advisor: Prof. Youngkeun Song

M.A., Landscape Architecture, Seoul National University — 2020.03 — 2022.02

B.S., Landscape Architecture, Kyungpook National University — 2011.03 — 2017.02

RESEARCH EXPERIENCE

Visiting Research Intern — Purdue University, FNR FUSE Lab (PI: Prof. Brady Hardiman) — 2023.12 — 2024.02
West Lafayette, IN, USA · forest structure & LiDAR

Visiting Research Intern — Virginia Tech (PI: Prof. Jaeyoung Ha) — 2024.07 — 2024.08
Blacksburg, VA, USA · landscape architecture

PEER-REVIEWED PUBLICATIONS

APA 7th · name in bold = author · † = first author · SCI then KCI, newest first.

1. Kim, J., Han, S., Yun, J., **Lee, S.**, & Song, Y. (2026). Ecological structures and terrestrial insect diversity across successional stages in abandoned paddy fields. *Agriculture, Ecosystems & Environment*, *399*, 110172. https://doi.org/10.1016/j.agee.2025.110172 · IF 6.4, Q1 (2026)
2. **Lee, S.†**, Ha, U., Lee, H., Choe, H., Kim, H., & Song, Y. (2026). Multi-scale forest typologies using novel vertical layer metrics from airborne LiDAR in temperate mixed forests. *Ecological Indicators*, *185*, 114798. https://doi.org/10.1016/j.ecolind.2026.114798 · IF 7.4, Q1 (2026)
3. Shao, J., Choi, D. H., Liu, J., Tian, X., Thapa, B., **Lee, S.**, Habib, A., & Fei, S. (2026). A three-stage framework for stand-level automated stem volume estimation in temperate forests using Mobile laser scanning. *Remote Sensing of Environment*, *335*, 115246. https://doi.org/10.1016/j.rse.2026.115246 · IF 11.4, Q1 (2026)
4. Yun, J., **Lee, S.**, & Song, Y. (2025). Assessing Corvus frugilegus (Rook) habitat preferences through flock-size-specific species distribution modeling using citizen science data. *Global Ecology and Conservation*, *63*, e03866. https://doi.org/10.1016/j.gecco.2025.e03866 · IF 3.4, Q1 (2025)
5. **Lee, S.†**, Song, Y., & Kil, S.-H. (2021). Feasibility Analyses of Real-Time Detection of Wildlife Using UAV-Derived Thermal and RGB Images. *Remote Sensing*, *13*(11), 2169. https://doi.org/10.3390/rs13112169 · IF 5.0, Q1 (2022)
6. **Lee, S.†**, & Song, Y. (2026). Forest-Type and Seasonal Analysis of GEDI-ALS Canopy Height Agreement and GEDI Structural Metrics (FHD, PAI): A Case Study of Forests in Gwacheon-si and Uiwang-si. *Journal of the Korean Society of Environmental Restoration Technology*, *29*(2), 21–40. https://doi.org/10.13087/kosert.2026.29.2.21
7. **Lee, S.†**, Jang, G., Song, Y., & Kil, S.-H. (2026). Comparing LST Estimation Methods under Matched Spatiotemporal Conditions in Low- and High-Rise Residential Areas: Satellite, Simulation, and UAV Approaches. *Journal of the Korean Society of Environmental Restoration Technology*, *29*(2), 41–59. https://doi.org/10.13087/kosert.2026.29.2.41
8. Jekal, G., Yang, Y., **Lee, S.**, & Song, Y. (2025). Non-destructive Carbon Storage Estimation of Salix Spp community In Wetlands Using LiDAR: A Case Study of Ungok Wetland. *Journal of the Korean Society of Environmental Restoration Technology*, *28*(6), 95–105. https://doi.org/10.13087/kosert.2025.28.6.95
9. Han, Y.-H., Shin, W.-H., Kim, J.-H., Kim, D.-H., Yun, J.-W., Yi, S.-Y., Kim, Y.-H., **Lee, S.**, & Song, Y. (2024). Diel Activity Patterns of Water Deer (Hydropotes inermis) and Wild Boar (Sus scrofa) in a Suburban Area Monitored by Long-term Camera-Trapping. *Journal of the Korean Society of Environmental Restoration Technology*, *27*(2), 55–65. https://doi.org/10.13087/kosert.2024.27.2.55

MANUSCRIPTS UNDER REVIEW

1. **Lee, S.†**, Choi, D. H., Song, Y., & Thorne, J. H. (2026). *Spaceborne LiDAR reveals post-fire vertical structural loss in a dense temperate Asian conifer plantation* [Manuscript under review]. Science of Remote Sensing.
2. **Lee, S.†**, Kim, H., & Song, Y. (2026). *Systematic bias in urban-forest vegetation coverage assessment: the influence of topo-edaphic conditions on discrepancies between ALS and field surveys* [Revise and resubmit, 1st round]. Geoscience Letters.
3. **Lee, S.†**, Kim, Y., Kim, D., Kim, H., & Song, Y. (2026). *Bi-temporal ALS assessment of vertical–horizontal forest structures and their structural association in a temperate urban forest* [Revise and resubmit, 2nd round]. Forest Science and Technology.
4. Kim, Y., **Lee, S.**, Shin, W., & Song, Y. (2026). *Integrating UAV-derived habitat metrics with movement persistence and species distribution models to characterize seasonal habitat use of invasive turtles in urban wetlands* [Revise and resubmit, 1st round]. Global Ecology and Conservation.
5. Jekal, G., Kim, Y. H., **Lee, S.**, Yun, J. W., Kim, D. Y., & Song, Y. (2026). *Monitoring transition-zone dynamics of a Phragmites communis–Suaeda japonica mosaic from multi-season Sentinel-2 in a coastal wetland* [Revise and resubmit, 1st round]. Journal of Coastal Conservation.

PATENTS (REPUBLIC OF KOREA)

11 patents — 7 registered, 4 pending. Highlights: drone-LiDAR wetland vegetation-structure determination (reg. 2025); spaceborne-LiDAR wildfire damage assessment (pending); tree-management platform (reg. 2025, sole inventor).

RESEARCH PROJECTS (SELECTED)

Integrated assessment tool for carbon reduction & ecosystem services — 2023 — 2027
Korean Ministry of Environment · Researcher · ₩1,579M (total)

Real-time positioning surveillance system for invasive species — 2021 — 2023
Korean Ministry of Environment · Project Manager · ₩1,208M (total)

Redundancy-based green-infrastructure technology for urban ecosystems — 2020 — 2022
Korean Ministry of Environment · Researcher · ₩927M (total)

Carbon storage/sink analysis & inventory for Gyeonggi-do by biotope — 2024 — 2025
Gyeonggi Research Institute · Researcher

Gwacheon-si urban ecological status mapping — 2021 — 2022

Gyeonggi Research Institute · Researcher
…and 7 additional national / municipal R&D projects (2018 — present).

CONFERENCE PRESENTATIONS & INVITED TALKS

International: 13 presentations (AGU ×4, ESA ×2, ICLEE ×2, World Forestry Congress, JCK) + 10 domestic (KOSERT / KSEE / KILA). Invited: NEF (Orlando, 2024), Asia Week (Fukuoka, 2024), SNU guest lectures (2025). Full list on request.

AWARDS & HONORS

Best Presentation Award, KOSERT (2022, 2025). Full graduate scholarships: Chung Mong-Koo Foundation (2022–), BK21 FOUR (2020–); Ilju Foundation undergraduate scholarship (2015–17).

TEACHING · SERVICE · PROFESSIONAL

Teaching: University Lecturer, Incheon National University (2024–25); 4 online GIS/QGIS courses (1,000+ students); 20+ invited workshops.
Service: Peer reviewer, Int. J. Applied Earth Observation & Geoinformation. Member: AGU, ESA, KOSERT, Ecological Society of Korea.
Professional: Founder, TREE:ID — urban tree-inventory & management SaaS startup (2023–25); Landscape Architect, SUNJIN E&A (2017–18).

TECHNICAL SKILLS

Programming: Python, R, Google Earth Engine
LiDAR / RS: ALS · GEDI · MLS, LAStools / PDAL, canopy & vertical-structure metrics; QGIS / ArcGIS, UAV & thermal imaging, ENVI-met
Machine learning: XGBoost, deep learning for forest & wildlife mapping
Languages: Korean (native), English (professional working)
Certifications: Engineer Forest, Engineer Landscape Architecture, Drone Pilot (Class 2)